

Student Answer Sheet

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Q1. Arteries carry blood away from the heart.

Example: Aorta carries oxygenated blood from left ventricle

Q2. Chloroplast is the organelle involved in photosynthesis.

It contains chlorophyll which captures light energy.

Q3. White blood cells protect the body against infections.

They fight against bacteria and viruses.

Q4. Transpiration is the process by which plants lose water.

It occurs mainly through stomata in leaves.

Q5. Nephron is the structural and functional unit of kidney.

Each kidney contains about 1 million nephrons.

Q6. Arteries carry blood AWAY from heart:

- Thick walls, no valves, carry oxygenated blood
- Example: Pulmonary artery, Aorta

Veins carry blood TOWARDS heart:

- Thin walls, have valves, carry deoxygenated blood
- Example: Vena cava, Pulmonary vein

Q7. Nephron Structure:

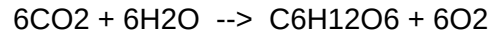
A nephron consists of:

- Bowman's capsule: Cup-shaped structure that filters blood
- Glomerulus: Network of capillaries inside Bowman's capsule
- Renal tubule: Reabsorbs useful substances

Q8. Photosynthesis:

The process by which green plants make their own food.

Balanced equation:



(In presence of sunlight and chlorophyll)

Q9. Translocation:

Transport of food (sucrose) from leaves to other parts.

Occurs through phloem tissue.

Food is transported from source (leaves) to sink (roots).

Q10. Breathing mechanism:

- Inhalation: Diaphragm contracts and flattens
- Intercostal muscles contract, ribs move outward
- Chest cavity volume increases, air rushes in
- Exhalation is the reverse process

Q11. Human Heart & Double Circulation:

The human heart has 4 chambers: 2 atria and 2 ventricles.

Double circulation means blood passes through heart twice.

1. Pulmonary circulation: Heart → Lungs → Heart
2. Systemic circulation: Heart → Body → Heart

This ensures efficient supply of oxygen to all body parts.

Q12. Transport of water and minerals:

- Water is absorbed by root hairs from soil
- Moves through root cortex by osmosis
- Enters xylem vessels
- Transpiration pull moves water upward
- Minerals are transported along with water